EMBEDDED SYSTEM FINAL PROJECT REPORT
DEPARTMENT OF ELECTRICAL ENGINEERING
UNIVERSITAS INDONESIA

BoboTekkom

GROUP 12

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PREFACE

Praise and gratitude be to God Almighty for all His grace and blessings, which have enabled us to complete the Embedded System Practicum Final Project report entitled "Bobotekkom: Smart Timer and PWM Control System" on time.

This report was compiled as a form of documentation and accountability for the project we have designed and implemented using the ATmega328P microcontroller, programmed entirely in AVR Assembly language. We would also like to express our deepest gratitude to our laboratory assistant, Christian H., as well as to all parties who have provided assistance and support throughout the development of this project.

We realize that this report may still have shortcomings; therefore, we highly welcome any constructive criticism and suggestions for future improvements.

Depok, Mei 17, 2026

Group 12

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CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

In modern daily life, lighting systems play an important role in supporting human comfort and productivity, especially during resting and sleeping activities. Many people often leave room lights turned on while sleeping, which can lead to unnecessary power consumption and discomfort during sleep. Therefore, an automated lighting control system is needed to improve user convenience while also increasing energy efficiency. The development of embedded systems technology enables the implementation of smart automation systems using microcontrollers. By utilizing timer-based programming and actuator control, a lighting system can automatically adjust its behavior according to user preferences. In addition, integrating features such as LCD display, buttons, PWM control, and serial communication allows the system to provide interactive feedback and better usability.

Based on these conditions, the project entitled **"BoboTekkom"** was developed as a smart sleep lighting system using the ATmega328P microcontroller programmed in AVR Assembly language. The system is designed to help users sleep more comfortably by automatically dimming or turning off the room light after a specified duration. The countdown timer is displayed on an LCD, while the user can configure the operating mode through push buttons. The project also implements several important embedded system concepts such as Timer Overflow Interrupt, Pulse Width Modulation (PWM), UART serial communication, and LCD interfacing. These features demonstrate the application of bare-metal programming techniques in controlling hardware resources directly at the register level.

Through this project, students are expected to gain deeper understanding of low-level microcontroller programming, interrupt-based multitasking, hardware interfacing, and real-time embedded system implementation.

1.2 PROJECT DESCRIPTION

BoboTekkom is an embedded-based smart lighting system designed to improve sleeping comfort by automatically controlling room lighting according to a specified countdown duration. The system uses the ATmega328P microcontroller as the main controller and is programmed entirely using AVR Assembly language. The main functionality of the system is to provide automatic light management during sleep. Users can start a countdown timer through push buttons, and the remaining time will be displayed on a 16x2 LCD module. When the countdown reaches zero, the system automatically changes the lamp state according to the selected mode, such as dimming the light using PWM or turning the light off completely.

The system also supports user feedback through UART serial communication, where countdown information and system status are transmitted to a serial terminal. This functionality helps monitor the internal operation of the system during execution and testing Several embedded system modules are integrated in this project, including Timer0 Overflow Interrupt for precise timing operations, PWM for brightness control, LCD interfacing in 4-bit mode for display output, and UART communication for serial monitoring. The implementation demonstrates direct register-level programming without relying on high-level libraries, making the project suitable for learning bare-metal embedded system development.

Overall, BoboTekkom combines automation, energy efficiency, and embedded system concepts into a practical application that enhances user convenience and demonstrates real-world implementation of microcontroller-based control systems.

1.3 OBJECTIVES

The objectives of this project are as follows:

1. To design and implement an embedded-based smart lighting system using the ATmega328P microcontroller
2. To develop a countdown timer system using Timer0 Overflow Interrupt for accurate time management without blocking the main system process
3. To implement Pulse Width Modulation (PWM) for controlling lamp brightness levels automatically

4. To provide user interaction through push buttons and visual feedback using a 16x2 LCD display
5. To implement UART serial communication for monitoring system status and countdown information
6. To apply bare-metal programming concepts using AVR Assembly language for direct hardware control
7. To improve user sleeping comfort and energy efficiency through automatic lighting management

1.4 ROLES AND RESPONSIBILITIES

The roles and responsibilities assigned to the group members are as follows:

Roles	Responsibilities	Person
Embedded System Programmer	Developing AVR Assembly program for PWM	Mohammad Ariq Haqi
Role 2	Designing LCD interfacing, UART communication, integrating modules, and device implementation	Nirmala Sari Zahiroh
Role 3	Developing AVR Assembly program for timer, interrupt handling, and countdown logic	Khalisa Zahra Maulana
Role 4	Coordinating project development and ensuring all modules work properly together	Yusri Sukur

Table 1. Roles and Responsibilities

CHAPTER 2

IMPLEMENTATION

2.1 EQUIPMENT

The tools that are going to be used in this project are as follows:

- Arduino Uno / ATmega328P Microcontroller
- 16x2 LCD Module (LM016L)
- Push Buttons
- LED Indicator
- Resistors (10kΩ)
- Breadboard and Jumper Wires
- UART Virtual Terminal
- USB Power Source
- Proteus Design Suite
- AVR Assembly Language
- Microchip Studio / AVR Studio

2.2 IMPLEMENTATION

The implementation of the BoboTekkom system was carried out by integrating hardware and software components into a single embedded system using the ATmega328P microcontroller. The system was programmed using AVR Assembly language to provide direct hardware-level control and improve understanding of low-level embedded programming concepts. The hardware implementation consists of several main modules, including push buttons, a 16x2 LCD module, LED output, UART communication interface, and timer-based control. The push buttons are used as user inputs for starting, resetting, and selecting the lighting mode. Meanwhile, the LCD module displays system status and countdown information in real time. The system utilizes Timer0 Overflow Interrupt to generate accurate timing operations without blocking the main program execution. The interrupt service routine (ISR) continuously updates the countdown timer every second by calculating the number of timer overflows. This method enables multitasking capability while maintaining efficient processor utilization. Pulse Width Modulation (PWM) is implemented using Timer1 to control the LED brightness. By adjusting the OCR1A register value, the system can produce different lighting intensities such as dim mode and off mode. In the implemented design, the LED brightness is initially set to approximately 50% duty cycle and automatically changes according to the timer status. UART serial communication is also integrated into the system for debugging and monitoring purposes. The remaining countdown time and system messages are transmitted through the serial terminal at a baud rate of 9600 bps. This feature helps users and developers monitor system behavior during operation and testing.

The LCD module is configured using 4-bit communication mode to reduce the number of required microcontroller pins. Commands and character data are transmitted through the data and control lines connected to PORTD and PORTB. The LCD displays important information such as system readiness, timer status, and lighting mode.Based on the Proteus schematic simulation, the system hardware includes an Arduino Uno board with ATmega328P, push-button inputs, pull-up resistors, LED output, virtual terminal UART interface, and an LM016L LCD module. The schematic implementation ensures proper communication between all hardware components and validates the overall system design. Fig 1 shows the complete hardware schematic of the BoboTekkom embedded system implemented in Proteus simulation software.

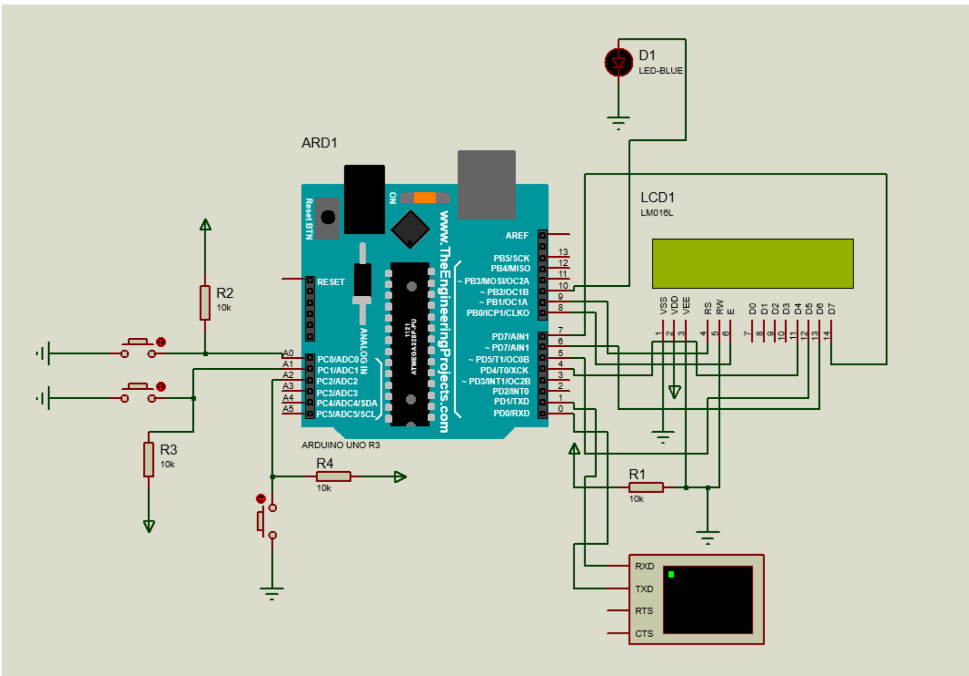

Fig 1. BoboTekkom System Schematic

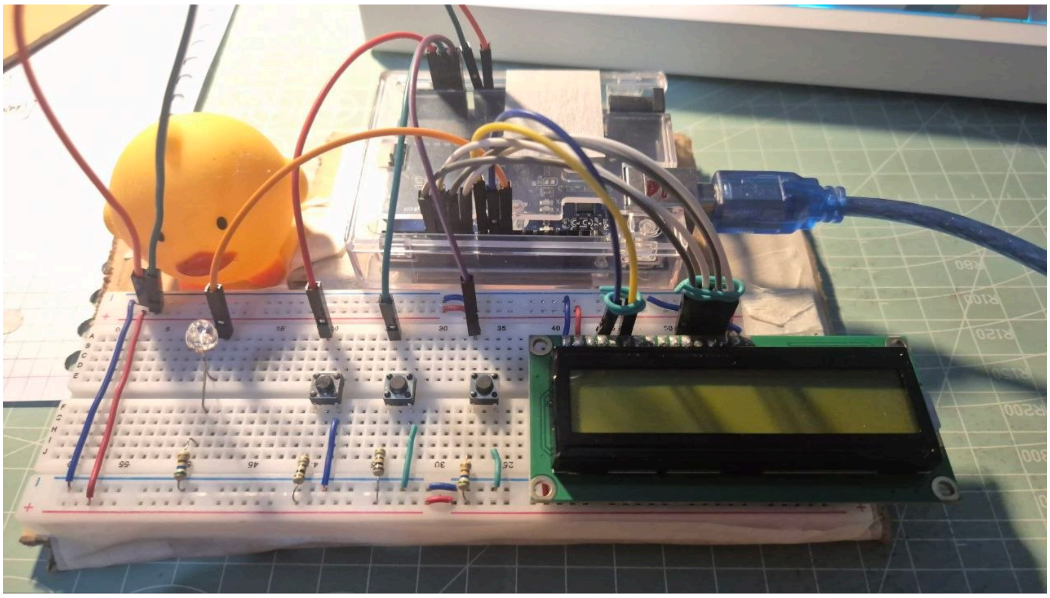

Fig 2. BoboTekkom Circuit

Overall, the implementation demonstrates successful integration of interrupt-driven timing, PWM actuator control, LCD interfacing, and UART communication within a bare-metal AVR Assembly environment. The project also shows how embedded systems can be applied to develop practical automation solutions for improving user comfort and energy efficiency.

CHAPTER 3

TESTING AND ANALYSIS

3.1 TESTING

3.1.1 PWM Toggle Testing via Button 3

The first test was conducted to verify the PWM brightness control behavior triggered by Button 3 (PC2). Upon system initialization, the LED was set to full brightness with OCR1B = 255. When Button 3 was pressed, the system jumped to the TOMBOL3_HANDLER routine, toggling register r25 from 0 to 1, which subsequently updated OCR1BL to 128, reducing the LED brightness to 50% duty cycle (dim mode). A second press restored r25 to 0 and OCR1BL to 255, returning the LED to full brightness.

Throughout this test, the global state registers r20 (state), r21 (tick flag), and r22 (timer expired flag) remained unaffected, as all OCR1B modifications were encapsulated within push/pop r16 sequences prior to and following each SFR access. This ensured that no register corruption occurred during the toggle operation. The UART serial monitor consistently outputted "light mode changed" upon each button press, confirming that the handler executed correctly and the state machine remained stable.

3.1.2 5-Minute Countdown Timer and LED Shutdown Testing

The second test was conducted to verify the complete timer countdown sequence and the automatic LED shutdown upon timer expiration. Upon pressing Button 1 (PC0), the system initialized r18 = 4 (minutes) and r19 = 59 (seconds), and transitioned r20 to state 1 (running), enabling the Timer0 overflow ISR to begin active timekeeping. The ISR decremented r19 on every 61st overflow as defined by OVF_PER_SEC, and upon r19 reaching zero, r18 was decremented and r19 was reloaded with 59. The serial monitor displayed a "sisa MM:SS" update each second, providing real-time confirmation of the countdown progress. Upon both r18 and r19 reaching zero, the ISR branched to the TIMER_HABIS label, setting r20 = 2 (completed state), writing OCR1BH = OCR1BL = 0 to fully extinguish the LED, and raising the expiration flag r22 = 1. The main loop subsequently detected r22 and invoked display_button_guide to update the LCD accordingly. Notably, all OCR1B write operations within the ISR were protected by push/pop r16, preserving the integrity of the previously saved SREG value on the stack and ensuring that the interrupt routine returned cleanly without corrupting the execution context of the interrupted foreground code.

3.2 RESULT

The testing conducted on the Bobotekkom system yielded successful results across all evaluated functionalities. The PWM brightness control operated as intended, with the LED responding correctly to each Button 3 press by toggling between full brightness (OCR1B = 255) and dim mode (OCR1B = 128). The timer countdown also functioned accurately, with the system decrementing the time registers every second through the Timer0 overflow ISR and automatically shutting off the LED upon expiration by setting OCR1B = 0.

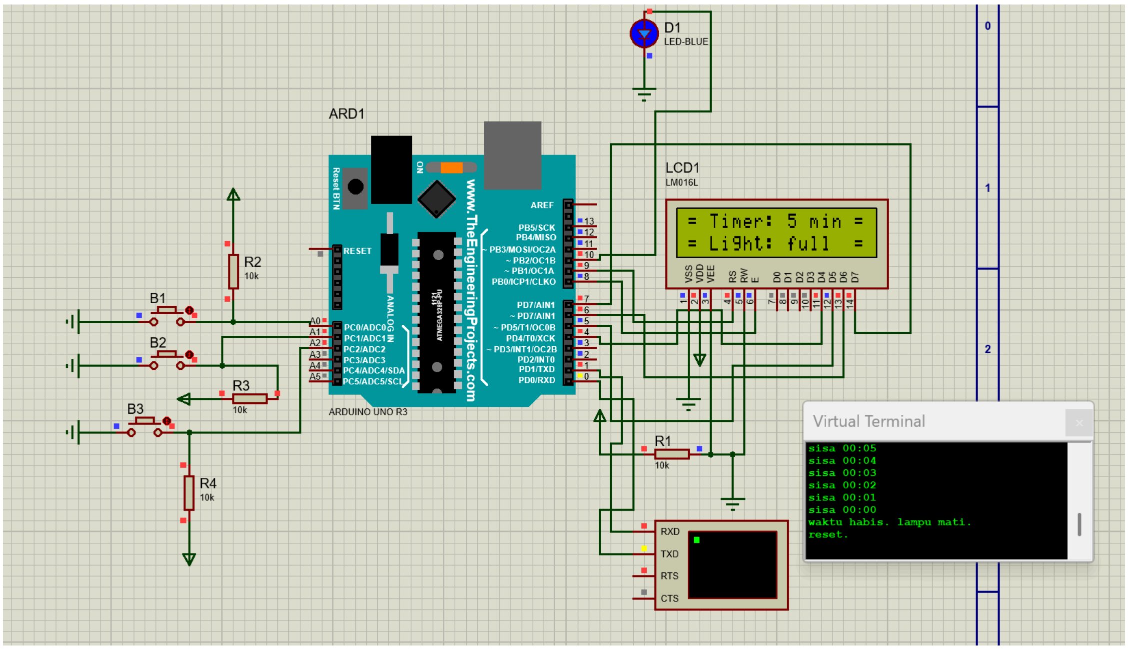

Fig 3. Testing Result A

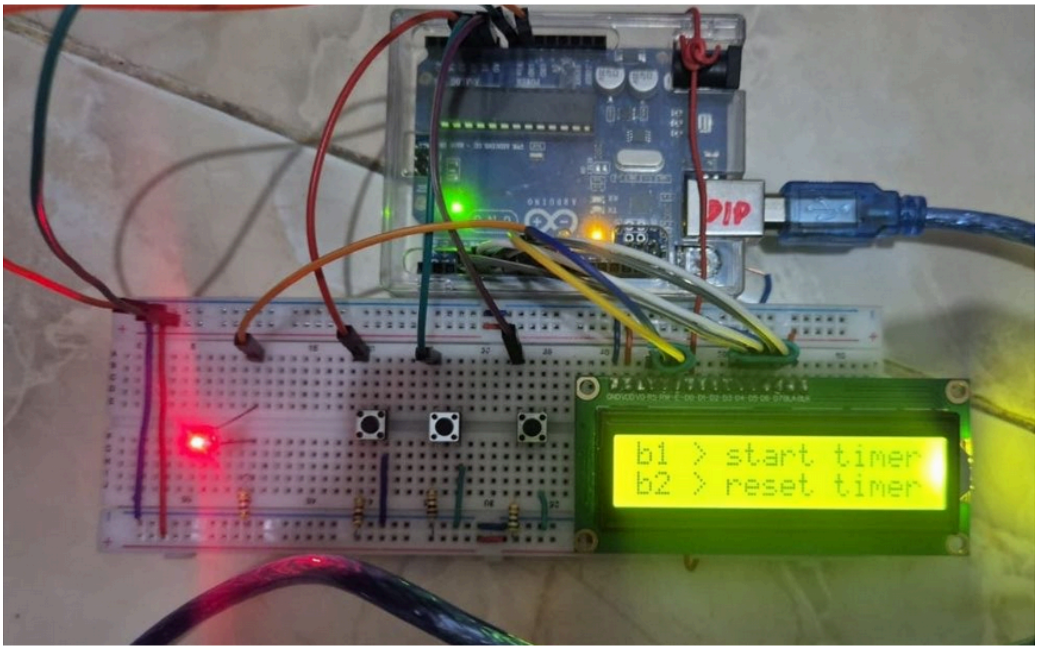

Fig 4. Testing Result B

Both the LCD display and UART serial monitor provided consistent real-time feedback throughout each test scenario, confirming that the state machine transitions between idle, running, and completed states were executed without fault. No register corruption was observed during operation, validating that the push/pop protection applied to all subroutines and ISR routines effectively preserved the integrity of the global state registers throughout the entire program execution.

3.3 ANALYSIS

The analysis covers two testing scenarios conducted to evaluate the system's core functionalities. In the first scenario, the PWM brightness control was verified by pressing Button 3 (PC2), which toggled register r25 between 0 and 1 to switch the LED between full brightness (OCR1B = 255) and dim mode (OCR1B = 128). Global state registers r20, r21, and r22 remained unaffected throughout the operation due to push/pop r16 protection applied around all SFR accesses. In the second scenario, the 5-minute countdown timer was initiated via Button 1 (PC0), setting r18 = 4 and r19 = 59 with r20 transitioning to state 1, enabling the Timer0 overflow ISR to actively decrement the time registers every 61 overflows until both reached zero, at which point the system branched to TIMER_HABIS, set OCR1B = 0 to extinguish the LED, and raised the expiration flag r22 = 1. In both scenarios, all critical register operations within the ISR and subroutines were protected by consistent push/pop sequences, ensuring no corruption of the global execution context throughout the program lifecycle.

CHAPTER 4

CONCLUSION

The design, implementation, and testing of the BoboTekkom system have been successfully executed using AVR Assembly language on the ATmega328P microcontroller through Arduino Uno R3. This project includes integration of multiple hardware and software peripherals, including 2x16 LCD interface, push button input, UART serial communication, and timer achieved using an optimized system by assembly. By utilizing the timer0 overflow Interrupt Service Routiner (ISR) calibrated to 61 overflow per second, the real-time countdown execution can be implemented without blocking the main program loop, thus enabling multitasking. Moreover, the implementation of Pulse Width Modulation (PWM) via Timer1 allows the system to modulate LED intensity between full brightness, dim mode, and automatic shutdown when the timer is 0. We also implemented context-saving with a push and pop mechanism to protect the important register and saving register for further use which will also eliminate risk of register corruption.

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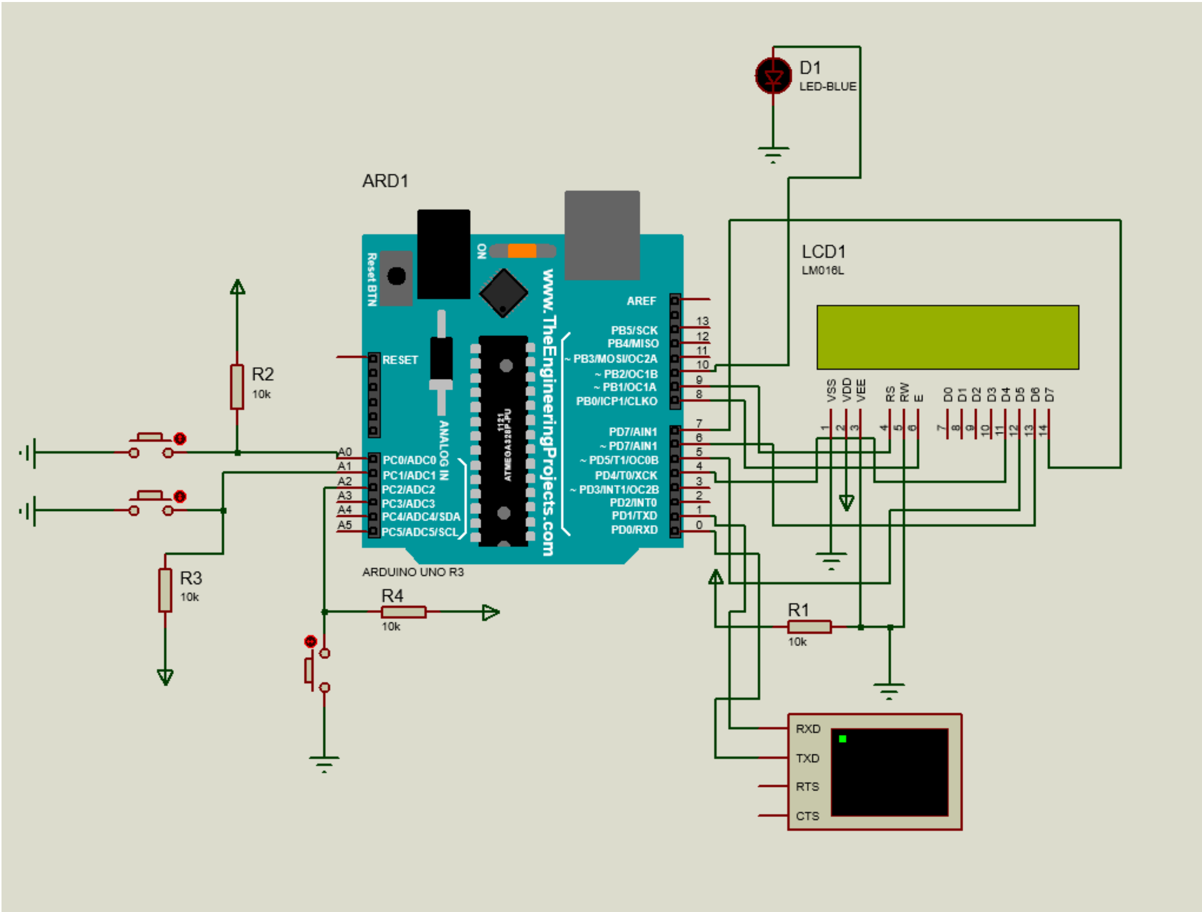
APPENDICES

Appendix A: Project Schematic



Appendix B: Documentation